

Gold Anchors License Nothing:

label-free auditing of LLM judge panels, an instrument that refuses,
and the measured failure of sanity checks

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Abstract

LLM judge panels are the de facto measurement apparatus of modern AI evaluation, and the de facto validation of those judges is a set of blatant sanity checks: exact-duplicate pairs, direct negations, honeypots—the crowdsourcing gold-question practice, transplanted. We built a label-free auditor for judge panels (anchored moment identification with refusal semantics and measured operating characteristics), sealed its own datasheet on simulation (nine of nine calibration gates), and then ran the measurement the practice assumes away: do blatant gold checks license the numbers a panel produces on real work? Across four constructed task families on a real correlated weak-judge panel, the answer is *no* at effect sizes that destroy every estimator tested: label-free coverage 0/15 in every family, while the same panel is flawless on the gold items. The failure is *silent* whenever the violation is smooth—our calibrated misfit flag fired on 15/15 replicates in one family and 1/15 in another. The repair is construction, not statistics: anchor strata drawn from the same generating process as the work items (“ladder anchors”) gave the correct answer in both of its possible directions—restoring calibrated coverage 13/13 when exactly one informative judge existed, and REFUSING (14/15 VOID) when honest anchors revealed that no judge was informative at all. Gold anchors certified garbage in both cases. A frontier-model panel audited at arm’s length under the same protocol was priced exactly, at ceiling, on every sheet including a deconfounded multi-hop task—locating the danger zone in the weak-to-mid judge tier, which is precisely the tier deployed for cheap bulk evaluation. The honest gold set does not merely fix the estimate; it tells you whether you have a panel at all. Two extensions close the constructed-corpus gap. On a real heterogeneous judge panel (a frontier API judge and local models) grading TruthfulQA, gold-anchor validity proves prevalence-dependent—it licenses correct numbers when the failure is rare and misses it by up to 0.249 as the failure becomes common, while ladder anchors cover the truth at every prevalence. And we sharpen the impossibility itself: two panels identical to 1.36×10^{-14} on unlabeled data—one sound, one unanimously wrong on a slice of items—are separated by a single known-negative (a $150.8\times$ likelihood ratio) and caught with better than 90% power by roughly 15 known-negatives under the standard test we ship. We then confirm the threshold on real model verdicts, where a detectable shared blind spot arises exactly when judges share a base model—a weak homogeneous panel is unanimously wrong on 0.64 of imitative falsehoods, and 3 known-label anchors detect it. The price of seeing a shared blind spot is not labels but a handful of known-negatives.

1 The impossibility, and what breaks it

Every ground-truth-free evaluator-accuracy estimator in the literature—Dawid–Skene (1979) through its modern weak-supervision descendants—requires conditional independence of judge errors given the true label, or a modeled correlation graph. Panels of LLM judges violate this by construction: shared base models, shared prompt templates, shared vulnerabilities. At three judges the confusion is exact—a correlated panel and an independent panel can induce identical joint verdict distributions on unlabeled data (exhibited constructively in our Stage-A witness; the equivalence is generic at that panel size). Known-label anchors break the impossibility: items whose truth holds by construction make per-judge error rates and error correlations observable, replacing the independence assumption with an anchor-exchangeability assumption. The entire content of this paper is that the exchangeability assumption is where the danger lives, that the standard practice violates it silently, and that an instrument can be built to price, flag, and refuse accordingly.

2 The instrument and its sealed datasheet

`styxx.anchors.audit_panel(V, neg, pos)` consumes an $n \times J$ verdict matrix and two anchor strata and returns either a refusal that names why (`VOID_PANEL__uninformative`; `VOID_ANCHORS__nonexchangeable` when the unclipped solution is impossible—never a clipped confident number) or an estimate wrapped in its own measured operating characteristics: a selection-aware bootstrap interval (the `sync` parameter activates only on evidence, and every bootstrap resample re-selects, so the interval prices selection uncertainty), a regime-keyed coverage note quoting measured rather than nominal rates, and a per-dataset misfit p -value from a parametric-bootstrap null with its power scope stated.

The characterization run sealed all nine calibration gates at the Stage-A design point: interval coverage 0.95 on clean validation panels (Wilson [0.888, 0.978]), 0.963 under a $\rho = 0.30$ shared failure factor, 0.912 and 0.938 at planted master-key rates 0.05 and 0.15; misfit false-alarm 0.090 on held-out clean replicates; deaf-panel VOID rate 0.967 under the plain informativeness gate and 1.000 under the noise-margin gate the instrument ships with; refusal false-fire 0/200; clean activation rate 0.020. Activation power against a true all-judge key: 0.30/0.7125/1.00 at wild rates 0.02/0.05/0.15—below roughly a five-percent key rate, absence of activation is not evidence of absence. Two prior interval constructions failed their coverage gates (0.835 and 0.850 on clean panels) and are on the public record as `CLOSED_NEGATIVE` with the diagnosed mechanism; the selection-aware construction is the one that sealed.

3 The kill: gold anchors on a real panel

The panel: four prompt-personas of a small instruction-tuned model (Qwen2.5-3B-Instruct, greedy), a real correlated panel of the kind deployed for bulk evaluation. The corpora: constructed-truth contradiction detection in four families (attribute facts, numeric consistency, temporal ordering, transitivity chains), each with a graded difficulty ladder, with truth held out from the judges. The gold anchors: verbatim pairs (negatives) and direct negations (positives)—the blatant ends an auditor can always construct.

Fifteen preregistered replicates per family. Label-free coverage of the true prevalence: 0/15, 0/15, 0/15, 0/15. Median audit errors 0.4658 (attribute), 0.2774 (numeric), 0.65 (temporal—one judge’s

anchor-to-organic false-fire gap reached -0.9992), 0.4444 (chains). The mechanism is measured, not argued: the judges catch every planted negation (anchor-to-organic sensitivity gap 0.0) and false-fire on 66 to 98 percent of REPHRASED-consistent organic items—“different words means different claims”—a failure mode a verbatim pair is structurally incapable of exhibiting. No estimator survives from the inside: majority vote and Dawid–Skene sit at median errors 0.55 to 0.60 , semi-supervised DS with the gold answers in hand 0.53 to 0.54 , and the anchored audit is least wrong and still hopeless. The deaf control—the same personas judging redacted content—was refused by the instrument in 90/90 replicates across the entire program: a content-free panel cannot extract a number.

The silence is the sharpest edge. The calibrated misfit flag—built precisely because the sealed datasheet showed smooth violations evade internal-consistency checks—fired on 15/15 wrong estimates in the numeric family, 14/15 in the attribute family, and 0/15 (temporal), 1/15 (chains) in the others: violations that bend every moment coherently leave nothing for a label-free check to see. Four families, one sentence: gold anchors license nothing, and sometimes nothing warns you.

4 The repair: ladders that estimate, and ladders that refuse

Anchor strata drawn from the same generator and difficulty mix as the organic items—still constructible without labels, because the constructor plants the truth—close the measured anchor-to-organic false-fire gap from 0.63 to 0.0322 (attribute) and to 0.0078 (chains). What happens next depends on what honest anchors reveal, and the instrument answered correctly in both directions. On the attribute family they exposed three of the four personas as uninformative (the plain gold anchors had certified all four) and the audit of the one real judge restored coverage 13/13 at median error 0.0626 . On the chain family they revealed that *no* judge was informative, and the instrument refused: VOID 14/15—an outcome named in the preregistration before the run, scored CLOSED_NEGATIVE under its frozen coverage clause, and reported as what it is. The honest gold set does not merely fix the estimate; it tells you whether you have a panel at all.

5 The frontier control, and a confound we caught ourselves

The same arm’s-length protocol (fresh-context judge sessions receiving only the task and the shuffled items; every verdict logged verbatim; the transcription trust boundary disclosed) was run on four personas of a frontier model. On every sheet—the attribute corpus, ordered multi-hop chains, and chains DECONFOUNDED after we caught our own generator emitting statements in rank order (making multi-hop queries positionally shortcuttable; the overclaim was demoted in a preregistration amendment before the corrected run)—the frontier panel scored 1.0 on all four personas, and the label-free audit priced it exactly: estimate equal to the held-out truth to machine precision, interval covering, every honesty mechanism idle. Twice more the four personas returned byte-identical verdict sequences, as they had on the first sheet: persona diversity is not error diversity; a frontier model in costumes is one judge, which is exactly the correlated world this instrument assumes. The danger zone for anchor non-transfer is the weak-to-mid judge tier.

6 In the wild: a real heterogeneous panel, and a prevalence-dependent licence

We ran the audit against its named residual—a real deployed panel, not a constructed one. A heterogeneous panel (a frontier API judge and two local instruction-tuned models: `gemini-2.5-flash-lite`, `Qwen2.5-3B-Instruct`, `Qwen2.5-1.5B-Instruct`; a fourth judge, `gemini-2.5-flash`, was voided at a 0.9304 parse rate—a thinking-model truncation, a harness limitation disclosed, not a judging result) graded TruthfulQA for factual correctness, whose imitative falsehoods are the panel’s real, un-injected failure mode. Positive class: the answer is incorrect. Fifteen preregistered replicates at each of three true prevalences; gold anchors (blatant true / absurd) versus ladder anchors (held-out real “Best Incorrect Answer”s). The design was adversarially reviewed before freezing (it caught a fatal we would otherwise have shipped—a single model in personas is byte-identical and would VOID both arms, reporting degeneracy as if it were a kill). The sensitivity control fired (every judge flagged the blatant-absurd positives at ≥ 0.95) and the panel was not rank-1 (cross-judge disagreement 0.69).

The result is a mechanism the constructed corpora could not show: **gold-anchor validity is prevalence-dependent**. As the true rate of incorrect answers rose across 0.30/0.50/0.70, the gold-anchored audit covered the truth in 14, 8, and 1 of 15 replicates, its median miss growing to 0.249; the ladder-anchored audit covered in 15, 15, and 15. The reason is measured: the judges’ false-negative rate on imitative falsehoods is 0.08 (flash-lite), 0.10 (Qwen-3B), and 0.72 (Qwen-1.5B), so blatant gold anchors—on which every judge is near-perfect—report an over-confident calibration that diverges from truth only once enough items are exposed to the weak judge’s real miss rate. Under the frozen forks this is a PARTIAL, not a kill: the strict bar (a confident gold miss of at least 0.15 with the ladder still covering, at two of the three prevalences) cleared only at 0.70, so no kill is claimed and the pattern is reported verbatim. The in-the-wild lesson is the paper’s thesis with a condition attached—gold’s licence is prevalence-conditional; the honest ladder’s is not, and it repaired the estimate in the wild (45/45) exactly as in simulation. The weak judge carries the failure, locating the danger empirically in the same weak-to-mid tier the constructed kill implicated.

7 The anchor threshold: the price of catching a shared blind spot

The impossibility in Section 1 is an impossibility about the unlabeled *marginal*; we make its escape quantitative. Construct two exchangeable three-judge worlds whose unlabeled verdict distributions are identical to 1.36×10^{-14} —one benign (independent judges, true prevalence 0.400) and one malign (a shared blind spot: all three judges unanimously wrong on 15% of the true negatives; true prevalence 0.304). They are genuinely different worlds—prevalence differs by 0.096—that no unlabeled statistic can separate: the majority-vote positive rate is 0.4056 in both, so every label-free estimator returns the same answer for the sound panel and the secretly-broken one.

A known-negative breaks it, because the *class-conditional*—what a known-negative samples—is exactly where the worlds diverge. The probability that a known-negative is unanimously mis-voted is 0.0010 in the benign world and 0.1508 in the malign one: a single unanimous-wrong known-negative carries a $150.8\times$ likelihood ratio for a shared blind spot. Under the standard most-powerful one-sided test on the unanimous-wrong count—which we ship as `styxx.anchors.blindspot_power` and `min_anchors_for_power`—roughly 15 known-negatives give better than 90% power to detect the malign world. An earlier, stricter rejection region put that count near 30, a valid lower bound

we now tighten; the same conservatism had assigned a single anchor power 0.0, contradicting its own $150.8\times$ likelihood ratio, whereas the standard test correctly makes one anchor already informative. The price of catching a unanimous-confident-wrong panel—the failure that unlabeled evaluation cannot see at any sample size, and the one that most endangers a safety eval—is not labels; it is a handful of known-negatives. (Exchangeable $J=3$; the mechanism is transparent, $LR = \tau/f^J$, so the numbers illustrate a clear law, not yet a general theorem—arbitrary J and a finite-anchor interval on the recovered prevalence are the next rungs.)

We then took the threshold out of simulation and onto real model verdicts (local Qwen judges grading TruthfulQA; imitative falsehoods as the candidate blind spot, blatant errors as the benign probe). A detectable all-judge blind spot appears exactly where the theory demands—a homogeneous panel of a weak model—and nowhere else: the unanimous-wrong rate on imitative falsehoods is 0.64 for one weak model run under three prompt paraphrases, 0.06 for a stronger model under the same paraphrases, and 0.02 for a heterogeneous cross-family panel. Heterogeneity and model strength each defend consensus; shared weakness does not. Where the blind spot fires, the panel agrees with itself 0.64 of the time while majority-voting the false answer correct 0.74 of the time—a self-agreeing panel that is confidently wrong, invisible to any consensus heuristic. Fed the measured error rates, `min_anchors_for_power` returns $K^*=3$ known-label anchors for 90% power, and on the real pool three anchors detect the blind spot at empirical power 0.959 against the predicted 0.953. The anchor threshold is not a simulation artifact; it is a usable instrument that bites precisely when judges share a base model—the correlated-failure danger this paper names throughout.

8 Foundation, scope, residuals

Every organic and anchor label in every scored corpus—36,720 items—was re-derived from item text alone by independent decision logic (a formal transitive-closure oracle for chains; slot, number, and form oracles for the rest): zero mismatches, zero undecidable. All gates preregistered with frozen bars; every miss reported verbatim; the program’s public record across this arc includes two interval constructions closed negative, a repair closed negative with its mechanism decomposed, a refusal resolution, and one self-caught confound demoted in writing before its corrected experiment ran.

Scope, stated plainly: one weak-judge base model carries the kill (an 8 GB device ceiling; the cross-vendor API arm is recorded as blocked on credits); all corpora are constructed; the frontier tier is measured only at ceiling because constructed ladders exhaust there. The named residuals—API-tier model generality, at least one in-the-wild evaluation setup where practitioners actually salt gold checks, and a naturalistic frontier-stressing ladder—bound the claim; they do not soften what is measured inside it.

9 Reproducibility

`pip install styxx` (7.26.0 or later); repository `fathom-lab/styxx`, `papers/anchored-validity/`. Preregistrations, harnesses, seeds, checkpoints, per-judge verdict receipts, oracle receipts, and OATH certificates for every finding are committed. The instrument that produced these numbers ships with its weaknesses printed on the label, and this document was number-verified by the same certifier that polices the rest of the program.

Every number in this document is bound to a committed receipt and machine-verified by `styxx.certify`; preregistrations were frozen and committed before every scored run, and the negative and refused outcomes above are reported under their frozen bars. Passing an audit is necessary, not sufficient: *the method can only void an eval, never bless one.*